Defense Study Plan

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Contents

1 Defense Study Plan **1**
- 1.1 Time budget (scale to what you have) . . . 1
- 1.2 Part 1 — Report re-read map . . . 2
  - 1.2.1 MUST . . . 2
  - 1.2.2 SHOULD . . . 2
  - 1.2.3 SKIM . . . 2
- 1.3 Part 2 — The stats gauntlet (give it a whole session) . . . 2
- 1.4 Part 3 — Reference refresh map . . . 3
  - 1.4.1 Tier A — must refresh (re-read the named parts) . . . 3
  - 1.4.2 Tier B — should refresh (confirm the exact terms, then state crisply) . . . 3
  - 1.4.3 Tier C — one-liner only . . . 4
- 1.5 Part 4 — Active-recall drills (do from memory, on paper) . . . 4
- 1.6 Part 5 — Slides and talk track . . . 4
- 1.7 Part 6 — Night before, one page . . . 4
- 1.8 Appendix — known wrinkle to be aware of . . . 5

1 Defense Study Plan

The actionable companion to `DEFENSE_PREP.md`. That document is the *what they will ask* (the question bank, R1–R12, the deep dive). This one is the *how to prepare* — which report sections to re-read, which references to refresh and to what depth, and what to drill until automatic. Cross-references like (R2) or (Tier1 Q4) point back into `DEFENSE_PREP.md`.

> **Section and table numbers below match the compiled report** (`Report/main.toc`, `main.lot`), not the markdown export. The markdown's chapter numbers are inflated by one (its "Chapter 4 / §4.5" is the real **Chapter 3 / §3.5**). The real chapter order is: 1 Introduction, 2 Background, 3 Forecasting Models and Evaluation, 4 System Implementation, 5 Forecasting-Model Results, 6 System-Evaluation Results, 7 Discussion, 8 Conclusion, A Glossary, B ML Test Score Worksheet.

Guiding principle: you are defending the **report**, in front of a stats / ML / reproducibility examiner who has not read the code. So re-read for three things at once — the argument spine, the exact numbers, and every place the prose claims more than the tables prove.

1.1 Time budget (scale to what you have)

A five-session plan. If you have fewer days, do Sessions 1, 2, and 5 and skim the rest.

1. **Session 1 — the spine.** Re-read the MUST sections below; be able to state the three RQs, the one-sentence contribution, and the headline result without notes.
2. **Session 2 — the stats gauntlet.** R1 and R2 specifically. This is where the likelihood times damage is highest for this examiner.
3. **Session 3 — reference refresh.** Tier A papers, targeted parts only.
4. **Session 4 — drills + a mock.** Active recall from memory, then a spoken run-through.
5. **Session 5 — slides, talk track, one-page cheat.** The night before.

1.2 Part 1 — Report re-read map

Priority tags: **MUST** (high-traffic, re-read closely), **SHOULD** (read once), **SKIM** (know it exists and the one-liner).

1.2.1 MUST

- **§1.3 Aim & research questions, §1.4 Contributions.** The spine. Walk away able to recite the three RQs verbatim-ish and the contribution in one sentence. Opening question and all closers. (Tier1 closers)
- **§3.1 Replication framing + §3.7 Deviations (Table 3.4).** The "faithful in specification, bounded in budget" line, and every row of the deviations table with its reason. (R7, Tier1 Q1)
- **§3.5 Evaluation protocol.** Re-read word for word — this is where the GW description lives, and it reads like the DM test. Know the daily loss-differential formula and the Newey–West HAC detail. (R1, R2)
- **§5.1 Headline accuracy + §5.2 Statistical significance.** The numbers (Tables 5.1/5.2) and the "tie." Note the report uses both "statistically tied" and the safer "neither can be shown more accurate" — rehearse leading with the second. (R1, drills)
- **§5.4 Comparison with published benchmarks.** The ranking inversion and its explanation. Find the sentence that states the budget cause and note how firmly it is worded. (R3)
- **§6.1 ML Test Score (Table 6.1) + §6.2 Reproducibility (Table 6.2).** The 2.5 (min-of-sections), the 66.7 %, the 7×10^{-6} re-run. (R4, R6)
- **§6.3 Drift + §6.4 Maturity placement (Table 6.3).** The 49/49 admission and the "Realised" labels; reconcile them. (R8)
- **§7.1 RQ1 (and §7.3 RQ3).** The synthesized answers; flag every "attributable" / "tied" you would soften out loud. (R1, R3, R8)
- **Wording traps to reconcile (read these exact claims and pre-write your concession):** the "complete audit trail from every forecast" sentences in §1.3 (Aim), §7.4 (Tying the threads, "any forecast can be reconstructed"), and Chapter 8 (Conclusion), against the 66.7 % in §6.2 (R4); the "daylight saving days are dropped" line in the Chapter 5 results intro and §3.5 (R9); "statistically tied" vs "indistinguishable" (R1).

1.2.2 SHOULD

- **§2.2 Lago benchmark, §2.3 Transformer, §2.4 MLOps practices, §2.5 evaluation methods.** The lit-review framing, so you can attribute every method you use to its source paper on demand.
- **§3.2 Data and features.** Be able to derive the feature count (see drills). The asinh rationale and the hourly-downsampling justification.
- **§3.3 Models (3.3.1 LEAR, 3.3.2 DNN, 3.3.3 Transformer).** Note §3.3.2's admission that the backtest DNN is single-seed (R5), and the Transformer's distance from the paper optimum in §3.3.3.
- **§3.4 Recalibration and backtesting.** The rolling-window ensemble, refit cadences, which windows each model uses.
- **§4.2 System at a glance + the P1–P9 practices (§4.3.1–4.3.9).** Be able to name all nine practices in order and the one artefact you map each to (Table 4.1 inventory, Table 6.3 maturity).
- **§7.2 RQ2, §7.5 Limitations.** The trade-off narrative and your own stated limits.

1.2.3 SKIM

- **§2.1** (Weron / Nowotarski background), **§4.3 P1–P9 prose** (know the rejected-alternative one-liners — Prefect over Airflow/Dagster, Loki over ELK, Compose over k8s, W&B over MLflow — but do not memorize the prose), **§4.4 cross-cutting**, the **appendices** (know the ML Test Score worksheet, Appendix B, exists and how the section sums add up).

1.3 Part 2 — The stats gauntlet (give it a whole session)

This is the highest-value hour of prep. For each, practice saying the answer aloud in two or three sentences.

- **"p=0.16 is not a tie."** Non-rejection vs equivalence; power at n≈ 187 against a ~3 % gap; the TOST/equivalence sentence said unprompted. (R1)
- **DM vs GW.** Write $\Delta_d^{A,B} = \phi' X_{d-1} + \epsilon_d$; explain that with X = a constant it reduces to the multivariate DM test (Lago eq. 15), which is what you compute; the full conditional GW adds lagged differentials. Know *why* GW exists (conditional predictive ability under recalibration). (R2)
- **Multiple comparisons.** No correction; why it does not bite (the load-bearing results clear any correction by orders of magnitude, the borderline cell is a non-rejection). (R1)
- **Newey–West.** Why a HAC standard error at all (autocorrelated daily loss differentials), and the lag rule $\lfloor T^{1/4} \rfloor$. (Tier2 stats)
- **Seed variance.** The conceptual answer (R5) — no experiment needed.

1.4 Part 3 — Reference refresh map

Refresh targeted parts, not whole papers. All PDFs are in `researh_material/`. Section numbers in this part are the **papers'** own numbers (not the thesis's). For the Tier B/C papers, the instruction is "re-read to *confirm* the exact terms" — do not walk in quoting a taxonomy from memory.

1.4.1 Tier A — must refresh (re-read the named parts)

- **Lago et al. 2021** (the anchor; 32 citations in your report). Re-read:
  - **§5.5.2 The Giacomini–White test** — eqs. 14 and 15. This is the source for the R2 answer; the constant-instrument reduction to DM is *in this section*. (§5.5.1 is the DM test itself.)
  - **§5.4 Evaluation metrics for point forecasts** — the three naive definitions and the explicit recommendation of $\hat{p}_{\text{naive},2}$ (price 7 days earlier). This vindicates your baseline; be able to quote it.
  - **§5.7 Ex-ante hyperparameter optimization** (the 1500-iteration TPE budget, the ex-ante principle) plus their forecasting-methods section for the DNN architecture, the eight tuned hyperparameters, and the four-run ensemble (§4.4). Backs R3 and the tuning-budget arithmetic.
  - **§5.8 Computation time** — the "more than 30 min / 1 h is unsuitable" rule of thumb, your analogue for the cost table.
  - **Table 3** (their rMAE values) — for the cross-paper ranking comparison.
- **Llorente & Portela 2025** (the Transformer; 12 citations). Re-read:
  - **§3.3 Retraining and Renormalization** — the key point that their model and DNN are evaluated *without retraining*, with a renormalization step instead. This is your sharpest RQ1 argument (their DNN was handicapped; yours recalibrates daily). (R3)
  - **Their Nord Pool validation table** (the HP grid, the n=512 embedding optimum) — for the Transformer-deviation answer.
  - **§4 results + their limitations** — they used the DM test only and flagged short test periods; context for why your significance-tested comparison adds something.
- **Breck et al. 2017** (ML Test Score). Refresh: the four sections (data, model development, ML infrastructure, monitoring), seven items each, the **min-of-sections** final score, and the **half-a-point-for-manual** rule. Backs R6 and §6.1.
- **Zarour et al. 2025** (the SLR; 16 citations — your second most-cited). Refresh: the **nine practices** (be able to define each in a sentence) and the **five maturity models**, especially why Google over Microsoft. Backs RQ3 and R10.

1.4.2 Tier B — should refresh (confirm the exact terms, then state crisply)

- **Gundersen & Kjensmo** (reproducibility in AI). Confirm their degrees/types of reproducibility, so you can correctly place your re-run as the weakest form (same code, same data, re-executed). (R6)
- **Runeson & Höst** (case-study method). Confirm the validity-threat categories (construct / internal / external / reliability) and the analytic-vs-statistical generalization distinction. Backs the case-study-method questions (Tier2 eval).
- **Mahadevan & Mathioudakis** (cost-aware retraining). Confirm the core result — triggered retraining pays off mainly when retraining is expensive — which is your justification for drift-as-monitoring-not-trigger. (R8, §6.3)
- **Diebold–Mariano 1995 / Giacomini–White 2006.** Do not read the dense econometrics; take the concepts from Lago §5.5.1–5.5.2. Be able to say in one line what each tests (unconditional vs conditional equal predictive ability). (R2)

1.4.3 Tier C — one-liner only

- **Weron 2014 / Nowotarski & Weron 2018** — the EPF and probabilistic-EPF reviews; one sentence each (background, and the CRPS/pinball direction for future work).
- **Uniejewski et al. 2018** — the asinh variance-stabilising transform; "defined for negative prices, unlike log."
- **Subramanya et al. 2022** — the DevOps-to-MLOps electricity-forecasting precedent (the Finnish FCR-N market on Kubeflow/Kubernetes); your delta is replicated benchmark models
  – significance tests + a quantitative maturity evaluation.
- **Google MLOps maturity model** — the three levels (manual, ML-pipeline automation, CI/CD-pipeline automation).

  Reference list is complete and verified against the report's citations. DORA is **not** cited in the final report (despite the PDF sitting in `researh_material/`), so do not prepare it.

1.5 Part 4 — Active-recall drills (do from memory, on paper)

If you cannot produce these without notes, you are not ready for that question.

- **Feature count.** Derive $10n + 7 = 247$ at $n = 24$ — name each block (4 price-lag blocks d-1/d-2/d-3/d-7, two exogenous at three lags d0/d-1/d-7, seven weekday dummies).
- **GW regression.** Write $\Delta_d^{A,B} = \phi' X_{d-1} + \epsilon_d$ and the constant-instrument reduction to DM.
- **Day-count funnel.** 220 calendar days → 204/201 (LEAR/DNN, DK1/DE) → 187/176 common, and why each drop happens (lag warmup, data gaps, the Transformer's 14-day sequence). Verified from the result CSVs.
- **rMAE.** The formula and why it is the headline (cross-period / cross-market comparability via the same-market naive).
- **The nine practices in order**, each with its one artefact (CI/CD → GitHub Actions; orchestration → Prefect; reproducibility → GIT_SHA + Compose; versioning → Git + migrations; collaboration → PRs + docs; train/evaluate → daily-forecast + reconciliation; metadata → ClickHouse tables + W&B; monitoring → Grafana/Loki/Prometheus + alerts; feedback → drift + retrain flag).
- **ML Test Score sums.** Data 2.5, model dev 3.0, infra 4.0, monitoring 4.0, min = 2.5, bound by data; why data binds.
- **Headline table.** rMAE per model per market and the key p-values (LEAR–DNN 0.16/0.19, vs Transformer ≤ 0.013, vs naive ≈ 0). The full table is in `DEFENSE_PREP.md`.

1.6 Part 5 — Slides and talk track

- **Pre-empt with a limitations slide before the conclusion** that names your own top four (the tuning budget behind RQ1 / R3, statistical power / R1, single-seed backtest / R5, the over-flagging drift detector / R8). Every question asked off your own slide is one you control.
- **Two whiteboard-ready explainers**, rehearsed: the feature vector to 247, and the GW/DM relationship.
- **The 60-second contribution statement**, memorized.
- **The standard closers**, written out: weakest point (pick one, own it), what you would do differently, who would use this and what is missing.

1.7 Part 6 — Night before, one page

Carry only this:

- The headline numbers table (rMAE, key p-values).
- The nine practices in order.
- The GW/DM one-liner and the feature-count derivation.
- The three RQ answers in one sentence each.
- "Weakest point" and "would do differently."

- The three concessions you have pre-written (audit trail → "every continuously-deployed forecast"; "attributable" → "most consistent with"; "tie" → "indistinguishable on this window").

1.8 Appendix — known wrinkle to be aware of

Two examiners, not one. Your supervisor (Finn Årup Nielsen) reads with an ML/stats/ reproducibility eye, which is why R1, R2, R6 are weighted highest. But a Danish defense also has an **external censor** who also reads only the report and whose background you do not control. Keep the system/MLOps narrative (RQ2, RQ3, the nine practices, the rejected-alternative justifications) sharp as well, in case the censor comes from that angle, rather than betting everything on the statistics questions.